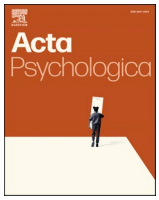
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Educational robotics and augmented reality for improving social skills of children with ADHD: A scoping review

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ABSTRACT

Attention Deficit Hyperactivity Disorder (ADHD) is the most common childhood neurodevelopment disorder that is characterized by symptoms of inattention, distractibility, impulsivity, and hyperactivity. To support children with ADHD to overcome developmental challenges, innovative technologies such as Educational Robotics and Augmented Reality have been introduced. The use of these technologies can contribute to enhancing cognitive and social skills. This scoping review aims to map how both technologies have been applied to support social skills in children with ADHD and summarize reported outcomes. A comprehensive literature search was conducted in Springer Link, IEEE, MDPI, ERIC, PubMed/MEDLINE, Web of Science, and PsycINFO for studies published in either Greek or English from 2015 to the present. The review identified seven studies meeting the inclusion criteria, with sample sizes ranging from 1 to 59 participants. Findings indicate that robots and AR systems can enhance children's engagement, attention, concentration, motivation, positive behaviors, and emotional recognition. Despite these promising results, the limited number of studies and small sample sizes highlight the need for further research to confirm these effects.

1. Introduction

1.1. Attention Deficit Hyperactivity Disorder (ADHD)

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder that emerges early in childhood and often persists across development, affecting multiple domains of functioning (Franke et al., 2018). Beyond its core symptoms of inattention, hyperactivity, and impulsivity, ADHD is consistently associated with significant difficulties in social functioning, particularly during the preschool and early school years, when peer interactions play a central role in socio-emotional development (Barkley, 2015).

Children with ADHD frequently experience challenges in forming and maintaining peer relationships, interpreting social cues, regulating emotional responses, and adapting their behavior to social contexts (Gardner & Gerdes, 2015). These difficulties often result in peer rejection, negative social feedback, and reduced opportunities for social learning, which in turn contribute to lower self-esteem and increased emotional distress (Meinzer et al., 2021). Consequently, the enhancement of social skills has been identified as a critical target for early intervention in children with ADHD.

ADHD is characterized by heterogeneous symptom profiles, commonly classified into inattentive, hyperactive/impulsive, and combined presentations (American Psychiatric Association, 2013). Regardless of subtype, impairments in executive functioning, emotion regulation, and behavioral inhibition are central features that directly interfere with children's ability to engage effectively in social interactions (Barkley, 2015).

Empirical evidence indicates that children with ADHD often struggle with turn-taking, maintaining reciprocal conversations, managing frustration, and responding appropriately to peers' emotional expressions (Christiansen et al., 2019; McKay et al., 2023). Difficulties in emotion recognition and regulation are particularly salient, with a substantial proportion of children with ADHD demonstrating deficits that exacerbate social misunderstandings and interpersonal conflict (Çiray et al., 2022). These challenges are especially pronounced in structured environments such as classrooms, where sustained attention, self-regulation, and cooperation are required.

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1.2. Social skills in children with ADHD: Conceptual framework

In the context of the present study, social skills are defined as the set of cognitive, emotional, and behavioral processes that enable children to perceive social information, regulate their responses, and engage in socially appropriate and goal-directed interactions (Elliott & Gresham, 2008). This definition is anchored primarily in the Social Information Processing (SIP) model (Crick & Dodge, 1994), and emotion regulation frameworks, which have been widely applied to understand social difficulties in children with ADHD (Eisenberg et al., 2010).

According to the SIP model, effective social behavior depends on a sequence of processes, including the encoding of social cues, interpretation of social intentions, generation of possible responses, and behavioral enactment (Crick & Dodge, 1994). Children with ADHD often exhibit disruptions at multiple stages of this process, particularly in cue interpretation and response selection, due to impulsivity and limitations in executive control (Barkley, 2015). Similarly, emotion regulation models emphasize the role of attentional control and inhibitory processes in managing emotional reactions during social interactions—domains in which children with ADHD commonly experience deficits (Christiansen et al., 2019).

Social skills in children with ADHD are typically assessed using a combination of standardized rating scales completed by parents and teachers, structured behavioral observations, and performance-based tasks that evaluate social problem-solving abilities, emotional understanding, and peer interaction patterns (Gresham et al., 2011). Establishing a clear operationalization of social skills is essential for the design, implementation, and evaluation of targeted interventions.

1.3. Educational robotics and augmented reality in general and special education

In recent years, educational technologies have gained increasing attention as tools for supporting learning and development in both general and special education contexts. Educational Robotics, in particular, has been shown to actively engage children in the learning process while fostering problem-solving, collaboration, and social interaction (Di Battista et al., 2020; Evripidou et al., 2020). In early childhood education, robotics-based activities have been associated with improvements in cognitive engagement, motivation, and foundational social skills (Zviel-Girshin et al., 2020).

Augmented Reality (AR) complements these approaches by overlaying digital elements onto real-world environments, creating interactive and potentially immersive learning experiences (Lorusso et al., 2018). AR may support attention, emotional engagement, and conceptual understanding, particularly for learners with special educational needs (Dhaas, 2024; Mikropoulos et al., 2020). Importantly, AR environments allow for the visualization and manipulation of social scenarios in a controlled and adaptable manner, thereby facilitating targeted practice of social behaviors.

The integration of Educational Robotics with AR has recently emerged as a promising instructional approach, offering multimodal interaction, immediate feedback, and opportunities for repeated practice in structured yet engaging contexts (Chen et al., 2020). These features align closely with the learning profiles of children with ADHD, who benefit from predictability, active participation, and reduced cognitive load.

1.4. Contribution of educational robotics in ADHD treatment

Non-pharmacological interventions that target social and emotional competencies are strongly recommended for young children with ADHD, particularly in educational settings (Danielson et al., 2018). Educational Robotics has been increasingly explored as a medium for such interventions, as robot-mediated activities can support attention regulation, sustained engagement, and social learning (Fridin & Yaakobi, 2011).

Social robots, in particular, provide consistent, non-judgmental interaction partners and can simulate everyday social situations, enabling children with ADHD to practice social behaviors in a safe and structured environment. Research suggests that interactions with robots may facilitate the development of empathy, emotional awareness, and socially appropriate behavior, while also supporting emotion recognition and self-regulation (Bakola & Drigas, 2020; Berrezueta-Guzman, Robles-Bykbaev, et al., 2021).

By combining Educational Robotics with Augmented Reality, interventions can further enhance the realism and flexibility of social scenarios, allowing children to engage in meaningful, context-rich interactions that directly target the social skill deficits associated with ADHD. This integrated technological approach represents a promising avenue for supporting social functioning in children with ADHD and constitutes the central focus of the present study.

The purpose of this study is to present a scoping review of the scientific literature examining the contribution of Educational Robotics and Augmented Reality to the enhancement of social skills in children with ADHD. Specifically, the review aims to identify existing intervention approaches, describe how they have been implemented, summarize reported outcomes, highlight gaps in the current literature, and propose directions for future research. The study is guided by the following research questions:

1. What types of interventions using Educational Robotics have been implemented to support social skills in children with ADHD, and what outcomes have been reported?
2. What Augmented Reality applications have been used to support social skills in children with ADHD, and what outcomes have been reported?
3. Have any interventions integrated both Educational Robotics and Augmented Reality, and what patterns of reported outcomes emerge from these studies?

2. Methods

2.1. Selection of publications

For this study, a scoping review was deemed the most appropriate method to systematically map and analyze the available literature regarding the use of Educational Robotics and Augmented Reality (AR) to enhance social skills in children with ADHD. Scoping reviews are particularly suitable for emerging fields, where research is limited and heterogeneous, allowing the identification of key concepts, research gaps, and directions for future studies (Munn et al., 2018).

To ensure a comprehensive and interdisciplinary coverage, the literature search was conducted across seven major databases: SpringerLink, IEEE Xplore, MDPI, ERIC, PubMed/MEDLINE, Web of Science, and PsycINFO. These databases were selected to encompass both educational/technological research and psychological/clinical studies. The search included studies published from 2015 to the present, reflecting the period when Educational Robotics and AR applications in education began to expand significantly.

The search was limited to publications in English and Greek. English-language studies were included to ensure international comprehensiveness, while Greek-language studies were intentionally included to capture local research and applications in educational and clinical contexts within Greece. This dual-language approach ensures that the review reflects both global evidence and region-specific insights, which are particularly relevant given that some interventions are developed and tested locally.

A systematic search strategy was applied using the following keywords: (“Educational Robotics” OR “Social Robotics” OR “Social Assistive Robotics”) AND (“Augmented Reality”) AND (“Attention Deficit Hyperactivity Disorder” OR “ADHD”) AND (“Social Skills”). This

strategy yielded 248 articles, which were imported into Mendeley reference management software to identify and remove duplicates. After deduplication, inclusion and exclusion criteria were applied. Inclusion criteria required studies to: (a) involve children with ADHD, (b) investigate the use of Educational Robotics and/or AR, (c) target the enhancement of social skills, and (d) be primary research studies published in peer-reviewed journals or conference proceedings. Exclusion criteria ruled out studies: (a) focusing on populations other than children with ADHD, (b) using technologies other than Educational Robotics or AR, (c) being reviews rather than primary research, or (d) involving participants over 18 years old. (See Table 1.)

Following these procedures, seven primary studies were included in the final analysis. The PRISMA 2020 framework was applied to ensure transparency and replicability, and a flow diagram (Flow Diagram 1) summarizing the selection process has been included in the manuscript. Detailed reasons for exclusion at each stage of the selection process are summarized in Supplementary Table 1, enhancing transparency and reproducibility of the review. Additionally, a Table 2 summarizing the characteristics of the included studies—including sample size, intervention type, duration, setting, outcome measures, and key findings—was added to facilitate comparison and clarity.

Table 1
Inclusion and exclusion criteria for the Scoping Review.

Category of Exclusion	Number of Records	Reason / Explanation
Duplicate records removed before screening	53	Exact duplicates identified and removed prior to title/abstract screening
Records removed for other reasons before screening	12	Non-relevant records identified during preliminary review of database exports
Records excluded during title/abstract screening	119	Did not meet inclusion criteria: • Population not children with ADHD • Technology not ER or AR • Not primary research (reviews, editorials, commentaries) • Target not social skills
Reports not retrieved	26	Full texts unavailable, inaccessible proceedings, or published in languages other than English/Greek
Reports excluded after full-text assessment	31	Specific reasons: • Reviews, not studies ($n = 4$) • Participants over 18 years old (n = 2) • Referred generally to Special Education ($n = 17$) • Referred to other technologies ($n = 6$) • Referred to cognitive skills, not social skills ($n = 2$)

2.2. Data synthesis and analysis

Data were analyzed using thematic analysis, which allows flexible and in-depth exploration of qualitative and quantitative findings (Braun & Clarke, 2006). This approach enabled the identification of common patterns, intervention characteristics, psychological mechanisms targeted, and observed outcomes related to the use of Educational Robotics and AR in children with ADHD.

Table 2
Characteristics of the included studies.

AUTHORS	TITLE	OBJECTIVE	TECHNOLOGY	SAMPLE	SOCIAL OUTCOMES	COGNITIVE/ BEHAVIORAL OUTCOMES
1. Amato et al. (2021)	Socially Assistive Robotics combined with Artificial Intelligence for ADHD	Recognition and recording of emotions Support and enhancement of attention during therapeutic sessions	Pepper Robot	5 children with ADHD (7–10 years old)	Enhanced collaboration and communication	Increased attention and engagement
2. Berrezueta-Guzman, Pau, et al. (2021)	Assessment of a Robotic Assistant for Supporting Homework Activities of Children With ADHD	Support of activities and homework for children with ADHD at home	Robotic Assistant "Atent@"	12 children with ADHD (6–12 years old)	Enhanced children's social behavior Increased participation	Improved task focus and adherence to instructions
3. Fernández-Batanero et al. (2022)	Educational Robotics and Attention to Diversity: A Case Study	Improvement of attention span in students with ADHD	WeDo 2.0 Robot	1 student with ADHD (11 years old)	Enhanced engagement in group activities; Some improvement in social interaction	Improved attention, concentration, fine motor, logical-mathematical skills
4. Peribañez et al. (2023)	An Experimental Methodology for Introducing Educational Robotics and Storytelling in Therapeutical Activities for Children with Neurodevelopmental Disorders	A guide for activity design and evaluation of their effectiveness A pilot study example using the Ozobot robot in a therapeutic context	Ozobot Robot	9 children with neurodevelopmental disorders (including ASD, ADHD, and DLD) (8–12 years old)	Improved collaboration and communication; Increased tolerance to distractions	Improved attention; Better task participation
5. Avila-Pesantez et al. (2018)	Towards the Improvement of ADHD Children through Augmented Reality Serious Games: Preliminary Results	Improvement of attention and concentration	AR "ATHYNOS"	11 children with ADHD (7–10 years old)	Increased social engagement; Improved cooperation	Improved attention, concentration frustration tolerance
6. Fraggista et al. (2023)	Exploring the use of augmented reality in teaching history to students with attention-deficit hyperactivity disorder	Investigation of the effectiveness of AR in teaching History	AR Application	10 children with ADHD enrolled in 1st year of secondary school	Improved participation and interaction in class	Improved attention, engagement, learning motivation
7. De Luca et al. (2024)	Serious Games for the Treatment of Children with ADHD: The BRAVO Project	Improvement of self-control, rule-following, attention, and concentration	AR BRAVO platform	59 children with ADHD	Improved self-regulation and cooperative behavior	Improved attention, rule-following, cognitive and behavioral skills

2.3. Study characteristics and data extraction

A standardized data extraction template was used to systematically collect key characteristics of the included studies. These characteristics included sample size, participant demographics, intervention type and setting, duration, outcome measures, and main findings. Only peer-reviewed primary studies were included to ensure methodological rigor. While no formal risk-of-bias or quality appraisal was conducted, the extracted methodological details allow readers to evaluate potential limitations regarding reliability, generalizability, and replicability of the findings. This approach is consistent with the scoping review methodology, which emphasizes mapping and describing existing evidence rather than formal quality assessment.

3. Results

The current scoping review analyzed seven primary studies examining the use of Educational Robotics (ER) and Augmented Reality (AR) to support social skills in children with ADHD, published from 2015 to the present. Four studies employed ER interventions, while three focused on AR applications. Across the included studies, sample sizes were small (ranging from 1 to 59 participants), the majority of participants were male, and interventions were conducted predominantly in therapeutic or alternative settings rather than mainstream classrooms. The outcomes assessed varied across studies and included social skills (e.g., collaboration, communication, emotion recognition), cognitive abilities (e.g., attention, concentration, self-regulation), and behavioral measures (e.g., adherence to rules, motivation for learning).

Regarding ER, two studies reported outcomes explicitly related to social skills, such as emotion recognition, positive social behavior, and collaborative engagement (Amato et al., 2021; Berrezueta-Guzman, Pau, et al., 2021). Interaction with social-assistive robots, including the humanoid Pepper, LEGO WeDo 2.0 kits, the Ozobot robot, and custom-designed robots such as Atent@, provided structured contexts in which children engaged with social behaviors. In these studies, ER provided a predictable and emotionally simpler interface than human interaction, supporting children in practicing social behaviors without fear or anxiety. For example, Amato et al. (2021) reported outcomes related to emotional recognition and engagement in five children interacting with Pepper, while Berrezueta-Guzman, Pau, et al. (2021) observed adherence to tasks and positive behaviors during home-based activities with 12 children using Atent@.

The remaining two ER studies primarily targeted cognitive outcomes but also reported secondary social-related outcomes. Fernández-Batanero et al. (2022) conducted a four-year intervention with a single 11-year-old student using LEGO WeDo 2.0, reporting engagement, attention, fine motor skills, logical-mathematical reasoning, and social participation. Similarly, Peribañez et al. (2023) implemented Ozobot activities with nine children with neurodevelopmental disorders (including ADHD, ASD, and DLD), observing engagement, attention, and task-related social interactions.

Regarding AR, all three studies primarily focused on cognitive skills but reported social-related outcomes. Avila-Pesantez et al. (2018) applied the AR game ATHYNOS with 11 children aged 7–10 years, reporting outcomes in attention, concentration, frustration tolerance, time management, and social participation during gameplay. Fraggista et al. (2023) evaluated AR-enhanced History lessons with 10 children in a classroom setting, reporting engagement, motivation, attention, and peer interaction. De Luca et al. (2024) developed AR-based serious games on the BRAVO platform, reporting outcomes related to self-regulation, rule adherence, attention, and social-emotional participation over a six-month period.

No studies were identified that integrated both ER and AR for children with ADHD, highlighting a gap in the literature. Some ER interventions combined robotics with other technologies, such as Virtual Reality, but these focused on other neurodevelopmental disorders, including Autism Spectrum Disorder (Lledó et al., 2024).

Across all included studies, several patterns were observed, including small sample sizes, pilot designs, and heterogeneous outcome measures, which limit generalizability. Multiple studies reported engagement, motivation, participation, and social skill–related outcomes; however, these findings are preliminary and context-dependent. Cognitive outcomes were often reported alongside social outcomes but were considered separately. Interventions mainly occurred outside mainstream educational settings, indicating a need for school-based research to assess ecological validity and practical applicability.

A summary table presents the key characteristics of the included studies, including sample size, setting, intervention type, comparison conditions (if any), outcome measures, key reported outcomes, and limitations. This synthesis maps similarities and differences across studies and provides a descriptive overview of the current evidence base regarding interventions for social skills in children with ADHD.

In conclusion, the results describe the types of ER and AR interventions used, the contexts and participants involved, and the outcomes reported. Evidence regarding potential effects is preliminary and highly context-dependent, underscoring the need for larger, controlled, and school-based studies, the use of standardized social skills instruments, and systematic examination of long-term outcomes.

4. Discussion

The present scoping review examined studies investigating interventions using Educational Robotics (ER) and Augmented Reality (AR) aimed at supporting social skills in children with ADHD. Seven primary studies were included, four employing ER and three employing AR. Overall, the evidence base is limited and preliminary, reflecting the emerging nature of research in this area. Across the reviewed studies, reported outcomes were related to attention, executive functioning, emotion regulation, social information processing, and social participation.

ER interventions were described as providing structured and engaging interactions that enabled children to practice social behaviors, such as turn-taking, cooperation, and emotion recognition, while sustaining attention and focus. AR environments were reported to offer immersive and controllable scenarios, enabling repeated observation and practice of social behaviors in safe, adaptive contexts. Both technologies were designed to target cognitive and emotional processes relevant to social competence, supporting children in processing social cues and regulating emotional responses, though the reported findings remain preliminary and context-dependent.

However, across the reviewed studies, several common intervention features were identified that may represent the “active ingredients” underlying the reported positive outcomes associated with both ER and AR technologies on social skill development in children with ADHD. Identifying these shared elements is important for understanding potential mechanisms of change and for informing the design of future interventions, regardless of the specific technology employed.

The first “active ingredient” identified is reduced social complexity; both ER and AR interventions simplify the social environment compared to naturalistic human interactions. Robots offer predictable, consistent responses without the subtle and rapidly changing cues characteristic of human communication. Similarly, AR presents controlled social scenarios where variables can be isolated and manipulated. For children with ADHD, who often struggle to process multiple social cues simultaneously due to attentional and executive function deficits (Barkley, 2015), this reduction in complexity may reduce cognitive demands and may allow focused practice on specific skills.

The second “active ingredient” relates to the predictability and consistency provided by both technologies. A hallmark of both technologies is the delivery of predictable, repeatable interactions. Unlike human partners, whose responses vary based on mood, context, and history, robots and AR systems respond consistently to the same inputs.

This predictability may help reduce anxiety associated with social uncertainty and may support the development of stable expectations about interaction sequences, which is a foundation for social learning that may be disrupted in children with ADHD due to difficulties with pattern recognition and anticipation in social contexts (Christiansen et al., 2019).

The third "active ingredient" concerns the immediate and consistent feedback provided by both technologies. Both ER and AR interventions typically provide immediate feedback on children's responses, whether through robotic reactions, visual cues, or game-based reward systems. Immediate feedback is particularly important for children with ADHD, who show reduced sensitivity to delayed reinforcement and benefit from proximal consequences that strengthen the association between behavior and outcome (Luman et al., 2005). This feature may support learning by helping to make the connection between social behavior and its effects explicit and timely.

The fourth "active ingredient" refers to high engagement through novelty and interactivity ensured by both technologies. The novelty of robots and immersive AR environments may capture and sustain attention, which provides a potential advantage for children with ADHD who struggle with sustained engagement in conventional learning contexts. The interactive nature of both technologies requires active participation rather than passive observation, which may enhance encoding and retention of social skills. Several reviewed studies reported increased motivation, engagement, and willingness to participate when children interacted with ER or AR systems compared to traditional activities (Avila-Pesantez et al., 2018; Fernández-Batanero et al., 2022).

The fifth "active ingredient" is the provision of a safe environment for practice and error. Both technologies create low-stakes environments where children can practice social behaviors without fear of negative social consequences. Mistakes made with a robot or in an AR scenario do not result in peer rejection, embarrassment, or damaged relationships. This psychological safety may encourage experimentation and risk-taking that children with ADHD might avoid in real social situations due to histories of negative peer feedback (Meinzer et al., 2021). Repeated practice in such contexts may contribute to building confidence and automaticity before skills are transferred to more complex real-world interactions.

Finally, both technologies can allow controlled pacing and adaptive difficulty. Unlike real-time human interactions that proceed at a pace determined by social partners, both ER and AR allow for pacing that can be adapted to the child's needs. AR systems can pause, repeat, or slow down scenarios, while robotic interactions can be structured to allow processing time. Some interventions also incorporated adaptive difficulty, adjusting task complexity based on child performance (De Luca et al., 2024). This scaffolding aligns with established principles of skill acquisition and may be particularly supportive for children with ADHD who require additional processing time due to executive function difficulties.

These shared features suggest that the potential value of ER and AR interventions may be related not only to the technologies themselves but also to how interventions incorporate these design principles. Future intervention design may benefit from explicitly incorporating these active ingredients, and research should investigate which features are most relevant for different social skill outcomes and for children with varying ADHD presentations.

4.1. Mechanisms linking technology features to social skill outcomes: A theoretical hypothesis

To advance understanding beyond descriptive synthesis, we propose conceptual models grounded in Social Information Processing (SIP) theory (Crick & Dodge, 1994) and emotion regulation frameworks (Eisenberg et al., 2010) that illustrate the potential mechanisms through which Educational Robotics and Augmented Reality may support social skills in children with ADHD. These models aim to clarify how each technology targets distinct psychological processes and SIP stages, thereby providing a theoretical foundation for interpreting the reviewed findings and guiding future intervention design.

4.2. Educational robotics pathway

Fig. 1 presents the proposed pathway through which ER interventions may support social skill development. The key features of ER, specifically predictable interaction patterns, embodied physical presence, reduced social complexity, immediate and consistent feedback, and a non-judgmental interaction partner, are hypothesized to influence several psychological mechanisms relevant to social functioning in children with ADHD.

Specifically, interaction with robots may reduce social anxiety by providing a safe practice environment free from the fear of rejection or negative evaluation. The novelty and interactivity of robots may help sustain attention and motivation, addressing core attentional difficulties associated with ADHD. Furthermore, the consistency of robotic responses allows for repeated behavioral rehearsal without the unpredictability inherent in human interaction.

From a SIP perspective, ER interventions may primarily support later processing stages, specifically **Response Selection** and **Response Enactment**. It should be emphasized that the idea that ER facilitates Response Enactment through embodied interaction remains theoretical and has not been empirically tested in the reviewed studies. Children can practice selecting appropriate social responses and executing social behaviors (e.g., turn-taking, cooperation, verbal communication) with a predictable partner, potentially reducing the cognitive load typically associated with real-time social monitoring. This may allow children to focus on skill execution rather than simultaneously managing the interpretation of rapidly changing social cues. However, ER may provide less direct support for earlier SIP stages (Cue Encoding and Interpretation), as robotic interactions, while valuable for practice, do not fully replicate the complexity and subtlety of human social cues.

4.3. Augmented reality pathway

Fig. 2 presents the proposed pathway through which AR interventions may support social skill development. AR technologies offer distinct features, such as immersive visual scenarios, controlled and repeatable practice opportunities, multimodal engagement, real-world overlay, and adaptive difficulty, that may target different psychological mechanisms than ER.

AR environments may enhance sustained attention through visual immersion and novelty. Critically, AR allows for controlled exposure to social cues without the time pressure of live interaction, enabling children to practice identifying and interpreting facial expressions, body language, and contextual information at their own pace. AR also provides opportunities for emotion regulation practice within safe, simulated scenarios where the consequences of errors are minimized.

From a SIP perspective, AR interventions may primarily support earlier processing stages, namely **Cue Encoding**, **Cue Interpretation**, and **Response Selection**. By presenting social scenarios in a controlled, repeatable format, AR enables observational learning and deliberate practice of cue identification and interpretation. Children can repeatedly view and analyze social situations, potentially supporting the development of schemas for recognizing emotional expressions and social intentions. However, AR may provide less direct opportunities for Response Enactment, as the virtual nature of AR interactions limits opportunities for embodied behavioral practice with a physically present social partner.

4.4. Proposed integrated model: Synergistic potential of combined ER and AR interventions

The complementary strengths identified above suggest a theoretical

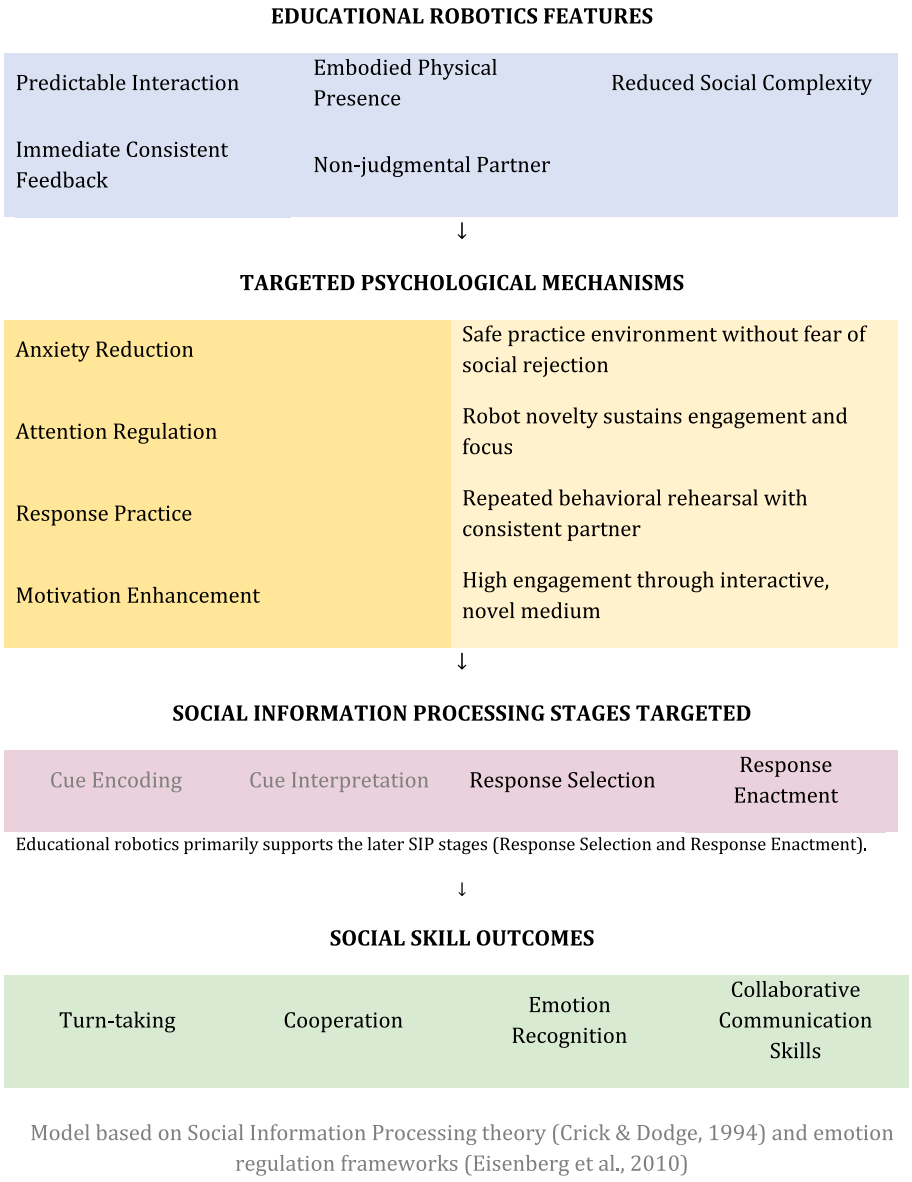


Fig. 1. Conceptual Model: Educational Robotics Pathway to Social Skill Enhancement in Children with ADHD.

rationale for integrating ER and AR within combined interventions. Fig. 3 presents a proposed integrated model illustrating how such a combination could potentially address the full SIP sequence.

In this framework, AR components would target earlier processing stages by providing immersive scenarios for observational learning, cue identification, and interpretation practice. ER components would target later processing stages by offering embodied, interactive opportunities for response selection and behavioral enactment with a predictable robotic partner. The integration point, namely Response Selection, represents a stage where both technologies may contribute, with AR supporting cognitive rehearsal of response options and ER enabling physical practice of selected responses.

This integrated approach may offer several potential advantages. First, it provides comprehensive coverage of the SIP sequence, addressing social skill development from initial perception through behavioral execution. Second, multimodal learning—combining visual-observational practice (AR) with embodied-interactive practice (ER)—may reinforce skill acquisition through complementary encoding pathways. Third, scaffolded complexity could be achieved by progressing from AR-based observation and interpretation to ER-based enactment, allowing children to build competence incrementally.

However, it is essential to emphasize that this integrated model represents a theoretical hypothesis requiring empirical validation. No studies included in the present review tested combined ER and AR interventions for children with ADHD. The proposed synergistic effects are extrapolated from the separate mechanisms identified in individual ER and AR studies. Future research should examine whether combined interventions yield additive or synergistic benefits compared to single-technology approaches, and identify optimal sequencing and dosage of each component.

4.5. *Limitations*

Several limitations were identified across the reviewed studies. Most interventions involved small sample sizes, pilot designs, and, in several cases, the absence of control groups, which limits the reliability and generalizability of findings. The measures employed to assess social skills were heterogeneous, including parent- and teacher-rated

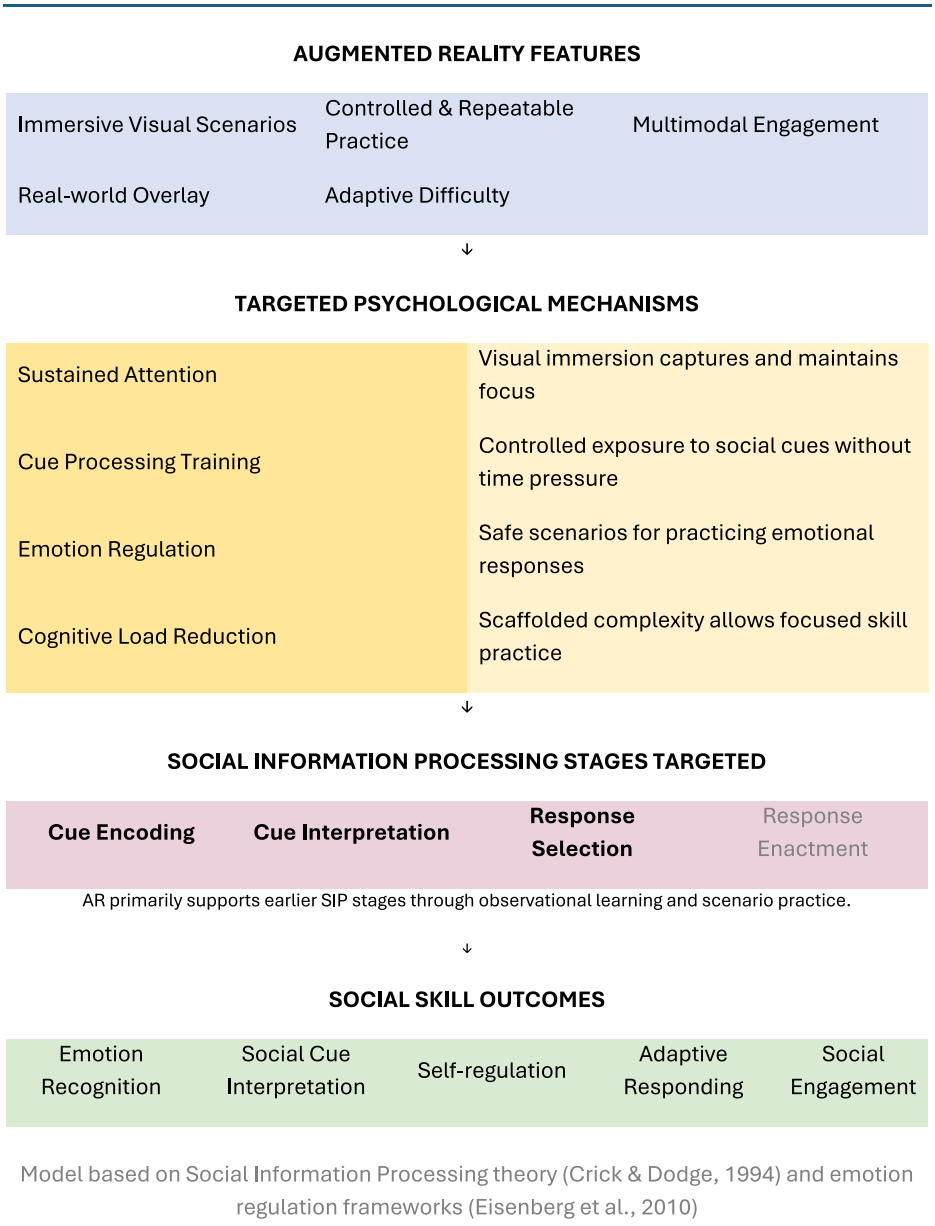


Fig. 2. Conceptual Model: Augmented Reality Pathway to Social Skill Enhancement in Children with ADHD.

questionnaires, observational tools, and performance-based tasks, complicating cross-study comparisons. Given the limited number of studies and small sample sizes, the reported outcomes should be interpreted cautiously. The current evidence provides preliminary insights and highlights directions for future research.

Additionally, interventions were primarily conducted in therapeutic or alternative settings, with only one study implemented within a mainstream educational context. This limits understanding of the ecological validity and practical applicability of these technologies in typical classroom environments.

Furthermore, although combined ER and AR interventions have been proposed as potentially beneficial, no empirical studies have yet tested this integration specifically for children with ADHD. Consequently, claims regarding synergistic effects of ER and AR remain hypothetical, highlighting the need for empirical validation.

4.6. Future research

Future research should focus on addressing these limitations through methodologically rigorous designs. Larger, controlled trials with diverse samples are necessary to strengthen the evidence base and allow for more robust conclusions. Employing standardized and validated measures of social skills will facilitate cross-study comparisons and ensure reliable assessment of intervention outcomes.

Investigating the integration of ER and AR in combined interventions represents a particularly promising avenue, as it may capitalize on the complementary strengths of both technologies to support attention, emotion regulation, and social learning. Furthermore, implementing interventions in mainstream classroom settings is crucial to evaluate feasibility, engagement, and broader educational impact.

Longitudinal research is also recommended to examine the sustainability of cognitive, emotional, and social outcomes over time. By addressing these research gaps, future studies may contribute to the development of more robust, ecologically valid, and scalable interventions for children with ADHD.

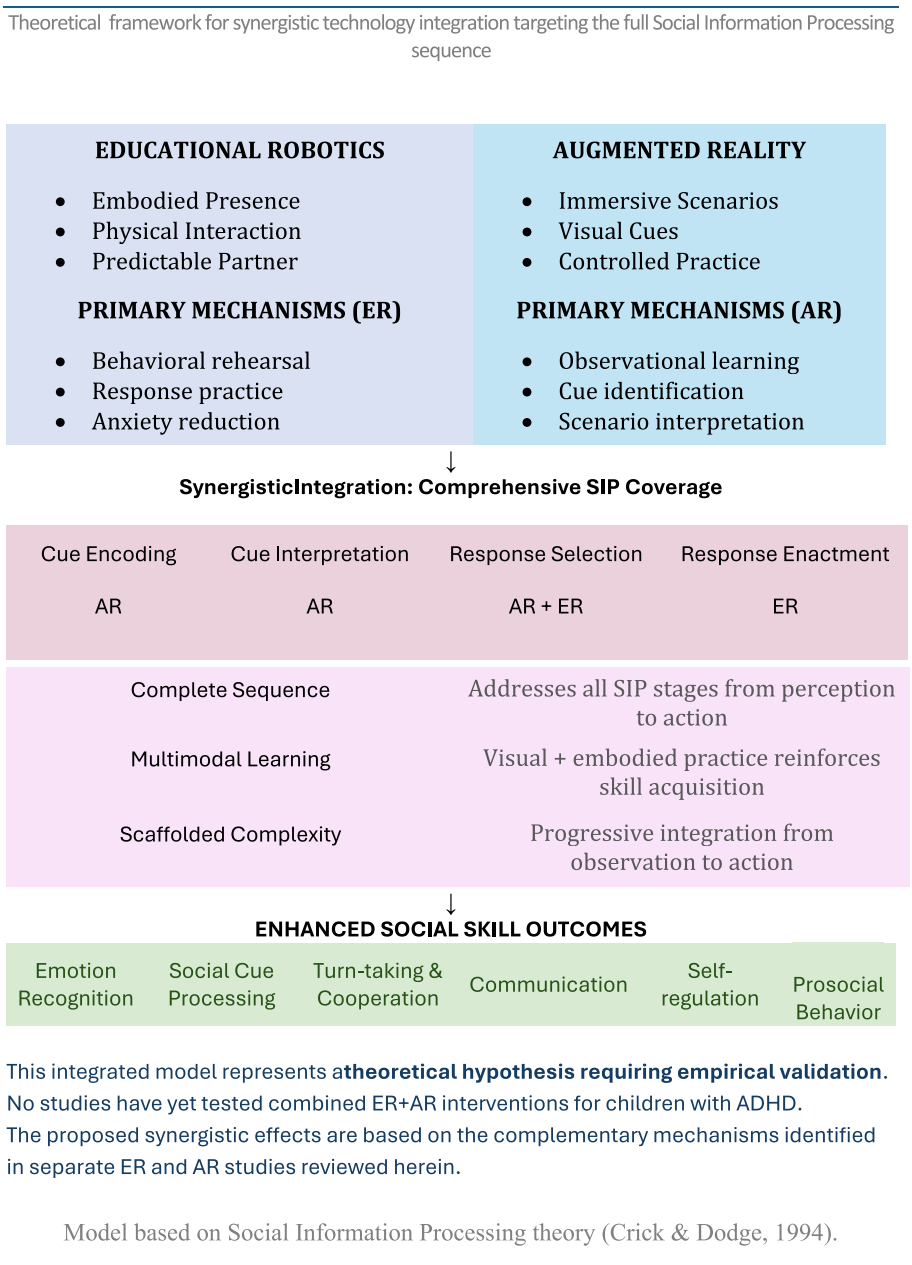


Fig. 3. Proposed Integrated Model: Combined ER + AR Intervention for Comprehensive Social Skill Support.

5. Conclusion

This scoping review synthesizes the current state of research on ER and AR interventions aimed at supporting social skills in children with ADHD. Despite the limited number of studies and the preliminary nature of the evidence, the reviewed studies reported outcomes related to attention, self-regulation, and social functioning. ER interventions were described as providing structured, interactive opportunities for practicing social behaviors, while AR was reported to offer immersive, adaptive scenarios that enable rehearsal of social skills in safe and controlled contexts.

The review highlights both the characteristics of the interventions and the gaps in the field. Few studies specifically targeted children with ADHD, no empirical research has yet tested combined ER–AR interventions, and most interventions were conducted outside mainstream classrooms. Additionally, heterogeneity in study designs, outcome measures, and settings limits comparability and generalizability.

By framing these interventions within the context of psychological models of social information processing and emotion regulation, this review provides a conceptual foundation for future research and a descriptive overview of intervention characteristics.

In conclusion, although research in this area remains at an early stage, the results describe the types of ER and AR interventions implemented, the contexts and participant characteristics, and the reported outcomes. Evidence regarding potential effects is preliminary and highly context-dependent, highlighting the need for larger, controlled, school-based studies, standardized social skills measures, and systematic investigation of long-term outcomes before drawing conclusions regarding efficacy.

CRediT authorship contribution statement

Maria Koukounara: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Korina Hatzinikolaou:** Writing – review & editing, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

AI-assisted technologies were used only to support translation and language polishing. All conceptual, analytical, and writing tasks related to the content of the manuscript were performed solely by the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.actpsy.2026.106882.

Data availability

No data was used for the research described in the article.

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